

A, B, O BLOOD GROUP

The ABO blood group system was discovered by the Austrian scientist **Karl Landsteiner**. ABO blood grouping is controlled by the gene I. The plasma membrane of the Red Blood Cells (RBCs) has sugar polymers that protude from its surface. These gene controls the kind of sugar protruding from RBC. The gene I has three alleles, I^A , I^B and i . The alleles I^A and I^B produce a slightly different form of sugar, while allele i does not produce any sugar.

On the basis of the presence or the absence of the antigen A or B, four types of blood groups are present, which are A, B, AB and O.

The Genetic Basis of Blood Groups in Human Population

Allele from parent 1	Allele from parent 2	Genotype of offsprings	Blood types of offsprings
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i	$I^A i$	A
I^B	I^A	$I^A I^B$	AB
I^B	I^B	$I^B I^B$	B
I^B	i	$I^B i$	B
i	i	ii	O

- For transfusion purpose the Rhesus factor and ABO grouping are the two most important compatibility factors to consider.
- An individual may be Rh^+ or Rh^- -factor. In simpler term, if an individual's blood type is A and positive for the Rhesus factor, then he or she is deemed to be ' A^+ '.

SEX DETERMINATION IN HUMANS

In humans, each cell contains 46 chromosomes (44 autosomes + 2 sex chromosomes). Male sex chromosome is XY, whereas female sex chromosome is XX. During gamete formation, in female, all gametes contain only one type of chromosome, i.e. X and in male, half of the sperm contain X-chromosome, while half contain Y-chromosome.

Thus, when a male gamete, i.e. sperm carrying X-chromosome fertilises an ova, the zygote develops into female and when a sperm carrying Y-chromosome fertilises an ova, the zygote develops into a male. Any increase or decrease in the number of sex chromosomes or autosomes causes genetic disorder.

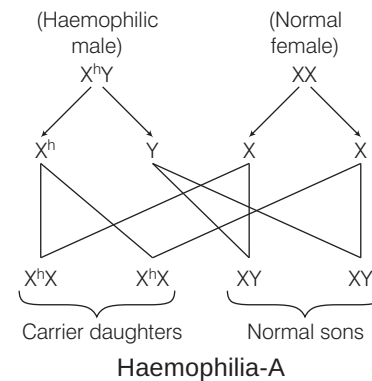
SOME HEREDITARY DISEASES

These are anatomical or physiological abnormalities present from birth, also called **congenital diseases**. These diseases can be caused by genetic mutations or environmental factors. **Mutation** is nothing but the sudden change in genetic make up due to addition, deletion or replacement of any gene.

Diseases by genetic mutation can be transmitted to next generation, whereas by environmental factors are non-transmissible.

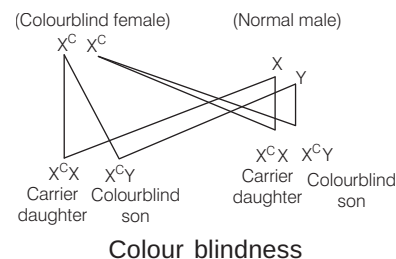
Haemophilia-A

- It is the most common X-linked recessive genetic disorder associated with serious bleeding.
- It is caused by a reduction in the amount and activity of coagulation factor VIII.



Colour Blindness

- It is an X-linked recessive genetic disorder, which results in the inability to perceive differences between some of the colours that others can distinguish.
- It does not mean not seeing any colour at all infact it leads to the failure in discrimination between red and green colour.



Cri-du-Chat Syndrome

- It is a rare genetic disorder due to a missing part of chromosome 5.
- It is characterised by having a **cat-like cry** of affected children.

Sickle-Cell Anaemia

- It is an autosome linked recessive trait that can be transmitted from parents to the offsprings, when both the partners are carriers for the gene.
- It is a genetic blood disorder characterised by an abnormal, sickle-shape red blood cells.
- It is caused by the substitution of glutamic (Glu) acid by valine (Val) at the sixth position of the β -globin chain of the haemoglobin molecules.

Down's Syndrome/ Trisomy-21

- It is a chromosomal disorder caused by the presence of all or a part of an extra 21st chromosome.
- It is associated with impairment of cognitive ability and physical growth and a particular set of facial characteristics.
- In this syndrome, the person exhibit **Mongolism**.

Klinefelter's Syndrome

- It is a condition, in which human males have an extra X-sex chromosome (44 + XXY).
- Effects are development of small testicles and reduced fertility.
- Such persons are sterile males or feminised males with undeveloped testis and some feminine characteristics like enlarged breasts, etc. They may be mentally retarded also.

Duchenne Muscular Dystrophy

- It is an X-linked disease associated with muscular dystrophy (mutations in the dystrophin gene).
- It is characterised by rapid progression of muscle degeneration, eventually leading to loss in ambulation, respiratory failure and death.

Turner's Syndrome

- It is a condition, in which human female has only one sex chromosome (XO).
- Effects are rudimentary ovaries and lack of secondary sexual character in women.

Patau Syndrome/Trisomy-13

- A syndrome, in which a patient has an additional copy of autosomal chromosome 13 due to a non-disjunction of chromosomes during meiosis.
- Its effects are mental retardation, cut mark in the lip.

Alkaptonuria

It is a rare inherited genetic disorder of phenylalanine and tyrosine metabolism.

Phenylketonuria

It is an autosomal recessive metabolic genetic disorder characterised by mutation in the gene for hepatic enzyme phenylalanine hydroxylase.

Progeria

Caused due to the point mutation in position LMNA gene replacing cytosine for thymine. Creating a truncated from progenin of prelamin a protein. Early aging, small face and jaw, pinched nose, loss of eye sight, hair loss are some of major symptoms.

Blue Baby Syndrome

It is used to describe newborns with cyanotic heart lesions. It is caused basically due to high nitrate contamination in ground water resulting in decreased O₂ carrying capacity of haemoglobin in babies leading to death. It is also called as **methaemoglobinemia**.

MOLECULAR BASIS OF INHERITANCE

Nucleic acid is the genetic material found in living beings. It was discovered by **Friedrich Miescher**. The term 'Nucleic acid' was given by **Altman**. Nucleic acids are polymers as nucleotides. Basically two types of nucleic acids are found in living systems, i.e. Ribonucleic Acid (RNA) and Deoxyribonucleic Acid (DNA).

Ribonucleic Acid (RNA)

RNA was the first genetic material to be discovered even before DNA evolved as a genetic material.

- The 2'-OH group of ribonucleotides is a reactive group that makes RNA, a catalyst. It is evident that the essential life processes such as metabolism, translation and splicing, etc., have evolved, around RNA being a catalyst was reactive and hence, unstable.
- Therefore, DNA has evolved from RNA with chemical modification that makes it more stable. RNAs are of three main types, i.e. messenger RNA (mRNA), transfer RNA (tRNA) and ribosomal RNA (rRNA), which participate in the process of transcription and translation also.
- RNA helps in the formation of protein, i.e. **protein synthesis** or **translation**.

Deoxyribonucleic Acid (DNA)

DNA acts as a genetic material in most organisms.

- It is a long polymer of deoxyribonucleotides.
- The number of nucleotides or base pair (bp) present in the DNA determines its length.
- DNA is made up of two polynucleotide chains and its back bone is constituted by sugar and phosphate. From this backbone the nitrogenous bases project inwards.
- The two strands of DNA are paired through hydrogen bonds (H-bonds) between bases.

Differences between DNA and RNA

DNA	RNA
It usually occurs inside nucleus and in some cell organelles like mitochondria and chloroplast.	Very little RNA occurs inside nucleus. Most of it is found in the cytoplasm.
DNA is the genetic material.	RNA is not the genetic material except in certain viruses, e.g. HIV, retrovirus.
It is double-stranded with the exception of some viruses like polyomavirus.	RNA is single-stranded with the exception of some viruses, e.g. double-stranded in T_2 , T_4 , T_6 bacteriophage, etc.
DNA shows regular helical coiling.	There is no regular coiling except in some parts of RNA.
It contains deoxyribose sugar.	It contains ribose sugar.
Nitrogen base thymine occurs in DNA along with three other adenine, cytosine and guanine.	Thymine is replaced by uracil in RNA. The other three are adenine, cytosine and guanine.
It replicates to form new DNA molecules.	It cannot replicate itself except in RNA-RNA viruses.
DNA controls heredity, evolution, metabolism, structure and differentiation.	RNA controls only protein synthesis.

Structure of a Polynucleotide Chain (DNA/RNA)

RNA and DNA both are comprised of polynucleotide chain. A nucleotide is composed of a nitrogenous base, pentose sugar and a phosphate group.

Nitrogenous Base

It is a nitrogen containing organic molecule having physical properties similar to a base.

- There are two types of nitrogenous bases:
 - Purines** (Adenine and Guanine)
 - Pyrimidines** (Cytosine, Thymine and Uracil)
- Adenine and thymine are bonded by double bond ($A=T$).
- Cytosine and guanine are bonded by triple hydrogen bond ($C\equiv G$).
- Out of pyrimidines, cytosine is common for both DNA and RNA, while thymine is present in DNA only.
- Uracil is present in RNA in place of thymine.

A Pentose Sugar

- Nitrogenous base is attached to the pentose sugar by N-glycosidic linkage to form nucleoside.
- Two types of sugars are present in RNA and DNA respectively, i.e. ribose in case of RNA and deoxyribose in case of DNA.

A Phosphate Group

When a phosphate group is attached to 5'-OH of a nucleoside through phosphodiester linkage, a nucleotide is formed.

Discoveries Related to Structure of DNA and Composition of DNA

- Friedrich Miescher** (in 1869), identified DNA as an acidic substance present in nucleus and termed it nuclein.
- James Watson** and **Francis Crick** (in 1953) proposed the double helix model of the DNA structure.
- Erwin Chargaff** observed that for a double-stranded DNA, the ratios between adenine, thymine and guanine, cytosine are constant and equal to one.

$$\frac{A+T}{G+C} = 1$$

Genetic Code

The relationship between the sequence of nucleotide on mRNA (i.e. messenger RNA) and sequence of amino acids in the polypeptide is called **genetic code**.

- In this nitrogenous bases in mRNA encode the information required for protein synthesis.
- The sequence of nitrogenous bases (nucleotides) in mRNA for coding a single amino acid is called as **codon**.
- A codon is triplet, i.e. it is formed of three nucleotide bases, specific, continuous, i.e. without comma or punctuation and universal in nature.

ORIGIN OF LIFE

The origin of life is considered a unique event in the history of universe. The study of history of life forms on the earth is called **evolutionary biology**. The important theories of origin of life are as follows:

Theory of Spontaneous Generation

- It states that life originated from decaying and rotting matter like straw, mud, etc., spontaneously, i.e. organism is formed automatically from non-living matter.
- Van Helmont** (1652) stated that young mice could arise from wheat grains, when these are put in the dark along with a moist shirt.

Theory of Biogenesis

- Theory of spontaneous generation (life) was rejected by **Francesco Redi** (1668), **Lazzaro Spallanzani** (1765) and **Louis Pasteur** (1861).
- Spontaneous generation of flies from rotting meat was disproved by F Redi.
- Spallanzani stated that air carried microorganisms.
- Pasteur is famous for 'Germ Theory of Diseases'.
- Theory of biogenesis states that the living things can only arise from previously existing living things.

Modern Theory (Oparin's Hypothesis)

- This theory is also known as modern theory or abiogenic origin or naturalistic theory or physiochemical evolution.
- According to this theory, life originated on earth after a long period of 'abiogenic molecular evolution'.
- It was hypothesised by **AI Oparin**. Oparin wrote a book '*The Origin of Life*' in **1936**.
- The first experimental support to Oparin-Haldane's theory of origin of life came from **Urey** and **Miller's** simulation experiment in 1953.
- They demonstrated clearly that the UV radiations or electrical discharge or combination of these can produce complex organic compounds from a mixture of CH_4 , NH_3 , H_2O and H_2 with ratio of methane, ammonia and hydrogen was 2 : 1 : 2 in the experiment.

- The most important condition for the origin of life is the presence of water.
- Hydrogen atoms were most numerous and most active in primitive atmosphere.

EVOLUTION

The term 'Evolution' has been derived from the Latin word *evolvere* (*e*-out; *volvere*-to roll) means to unfold or to unroll to reveal hidden potentialities. Evolution is a process by which different kinds of living organisms develop from earlier forms.

Geological Time Scale and Events in the Evolution

By studying fossils occurring in different strata of rocks, geologists are able to reconstruct the time and course of evolutionary change.

Time Scale of Earth

Era	Period Age in million years from present	Epoch Age in million years from present	Fauna (Animals)	Flora Plants
Cenozoic (Era of modern life) Age of mammals and angiosperms	Quaternary 0-2	Recent (Holocene) (0.01)	Modern man dominant; Modern mammals, birds, fishes, insects.	Rise of herbaceous plants, decline of woody plants.
		Pleistocene 2	Extinction of great mammals; Humans appeared evolution of human society and culture.	
	Tertiary 2-65	Pliocene 6-7 (Age of mammals)	Formation and adaptive radiation of modern mammals.	Adaptive radiation of flowering plants.
		Miocene 26	Mammals at peak, first man-like apes formed.	
		Oligocene 38	Rise of first monkeys and apes.	
Mesozoic (Era of medieval life) Age of reptiles and gymnosperms		Eocene 54	Diversification of placental mammals, origin of horse.	Angiosperm dominance.
		Palaeocene 65	Rise of first primates.	
	Creataceous 135		Extinction of dinosaurs and toothed birds; Rise of and placental mammals.	Dominance of angiospermic plants.
	Jurassic 145 (Age of reptiles)		Origin of toothed birds (first birds) dinosaurs dominant.	Origin of angiosperm. Dominance of gymnosperms.
	Triassic 225		Origin of dinosaurs and mammals.	Abundance of gymnosperms.
Palaeozoic (Era of ancient life)	Permian 280		Extinction of trilobites. Origin of mammals-like reptile (therapsids) and most modern orders of insects.	Origin of conifers.
	Carboniferous 350 (Age of amphibians)		Origin of reptiles and winged insects. Amphibians dominant.	Abundance of tree ferns, forming coal forests.
	Devonian 400 (Age of fishes)		Origin of amphibians; fishes abundant spiders appeared.	Earliest mosses and ferns. First seed plants appeared.
	Silurian 440		Origin of jawed fishes and wingless insects; earliest coral reefs.	Earliest spore-bearing plants.
	Ordovician 500 (Age of invertebrates)		Origin of vertebrates (Jawless fishes); invertebrates abundant, also called age of giant molluscs.	
Proterozoic (Era of early life)	Cambrian 570		All invertebrate phyla established; origin of trilobites.	
	1000, 2000, 3000		Ancient metazoans (sponges, cnidarians), primitive eukaryotes; scanty fossils (prokaryotes).	
Archaean (Era of invisible life)	3500		Origin of life; no recognisable fossils.	
Azoic (Era of no life)	4600		Origin of solar system; no life.	

Evidences of Organic Evolution

- **Homologous organs** are similar in basic structure and origin, but dissimilar in functions, e.g. wings of bat, cat's paw, front foot of horse, human hands and wings of birds are homologous organs.
- **Analogous organs** are the organs similar in shape and function, but their origin, basic development plan are dissimilar, e.g. wings of insects, birds and bats are analogous organs. Analogous organs are also called as **homoplastic organs**.
- **Vestigial organs** are degenerated, non-functional organs, which were functional earlier. Human body has been described to possess about 90 vestigial organs.
- Some of these are muscles of ear pinna, canine teeth and third molar teeth, body hairs, vermiform appendix (degenerated terminal part of caecum) nictitating membrane of eye, caudal vertebrae (coccyx or tail bone), etc.
- **Atavism or reversion** is the sudden reappearance of some ancestral features. Appearance of thick body hair, large canines, monstrous face, short temporary tails, additional pairs of nipples, etc., are examples of atavism.
- Connecting links are organisms that have features of two existing groups of organisms. These also provide an evidence for evolution.

Organism	Connecting link between
Virus	Living and non-living
<i>Euglena</i> (Protozoa)	Plants and animals
<i>Proterospongia</i> (Protozoa)	Protozoa and Porifera
<i>Peripatus</i> (Arthropoda)	Annelida and Arthropoda
<i>Neopilina</i> (Mollusca)	Annelida and Mollusca
<i>Balanoglossus</i> (Chordata)	Non-chordata and Chordata
<i>Dipnoi</i> (Lung fish)	Pisces and Amphibia
<i>Archaeopteryx</i> (Aves)	Reptiles and Aves
<i>Prototheria</i> (Mammalia)	Reptiles and Mammalia

Theories of Evolution

Evolution is generally progressive and variation is most important requirement for it. The ultimate source of organic variation is mutation (sudden change in genes). These are some important theories of evolution:

Lamarckism

The French biologist **Jean Baptiste de Lamarck** (1744-1829) suggested a complete theory of evolution. Lamarckian theory is also known as 'Theory of Inheritance of Acquired Characters' or 'Theory of Use and Disuse of Organs'.

- (i) Lamarck's theory was published in his book '*Philosophie Zoologique*' in 1809.

- (ii) Lamarck arranged his theory in four postulates.
 - Internal forces tend to increase size of the body.
 - Formation of new organs is the result of the need.
 - Development of organs is based on the continuous use and disuse.
 - All changes acquired by the organism are transmitted to offspring by the process of inheritance.
- (iii) Some examples of uses and disuses of organs are:
 - Webbed feet of swimming birds.
 - Rudimentary eyes of cave dwellers.
 - Elongated limbless body of snake.
 - Long thin neck in giraffe.
 - Vestigial organs of living animals.

Darwinism

Charles Darwin (1809-1882) and **Alfred Russel Wallace** (1823-1913) proposed, the 'Theory of Natural Selection'.

Darwin's theory is based on five principles:

- (i) Over production is shown by every organism.
- (ii) Organisms show struggle for existence.
- (iii) Struggle for existence leads to variations and their inheritance.
- (iv) The variations in the organisms lead to the survival of the fittest.
- (v) Natural selection and species formation is due to variations.

One major criticism against Darwin's theory was his failure to give an explanation for variations.

Mutation Theory

Hugo de Vries (1845-1935) gave much importance to the discontinuous variation or saltatory variations and proposed that new species arise not by the accumulation of minor and continuous variation through the natural selection, but by saltatory variations that appear suddenly for, which he coined the term 'mutation'.

- (i) After that he proposed the 'theory of mutation'.
- (ii) Mutations are discontinuous variation.
- (iii) These are due to changes in chromosomes, genes and DNA. These changes may or may not be inherited.

Synthetic Theory

The modern theory of origin of species or evolution is known as **modern synthetic theory** of evolution.

- (i) Initial basis of synthetic theory was given by **Dobzhansky** (1937). Modern synthetic theory of evolution was designated by **Huxley** in 1942.
- (ii) According to it the five basic factors are:
 - Gene mutation
 - Changes in chromosome structure and number
 - Genetic recombinations
 - Natural selection
 - Reproductive isolation

First three factors are responsible for genetic variability.